Ateeb Rehman

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Education

Imperial College London Oct 2022 – Jun 2027 (expected)

MEng Mechanical Engineering with a Year in Industry: Expected First

- Key modules: Business Economics (82%), Finite Element Analysis (77%), Fluid Mechanics 3A (77%).

Langley Grammar School Sep 2015 – Jul 2022

A-Levels: Maths (A), Further Maths (A*), Physics (A*)*

- Prefect: led a class of Year 10 students in weekly sessions focused on developing presentation and communication skills.
- Logistics Lead: managed logistics for weekly Jummah prayer sessions as attendance grew from about 20 to 200 over 7 years.

Professional Experience

Caterpillar Inc. | Simulation Engineer — Thermofluids Jan 2026 – Jul 2026

- Delivered a breather design study for an internal customer 2 weeks ahead of schedule, comparing 5 designs across 6 operating conditions; asked to take on the follow-on analysis after the selected design progressed to the next stage.
- Showed that a new engine in development met air–fuel ratio targets at all altitudes up to 4,500 m with a derate applied, providing evidence for a turbocharger sourcing review.
- Presented progress weekly to a 6-person engineering team in China, using their feedback to steer next steps.
- Validated a design combining two coolant pipes into one to reduce assembly complexity, weight and cost, confirming it met flow and pressure requirements with an estimated 10% lower pressure loss.
- Organised weekly Jummah prayer sessions for the site's minority Muslim community, coordinating colleagues across multiple teams to align with everyone's schedules.

Caterpillar Inc. | Test Facilities Engineer Jul 2025 – Dec 2025

- Built a Power BI asset health dashboard from scratch covering over 5,000 assets, adopted by the whole Test Facilities Emissions department to speed up decision-making; presented it to the department's senior leadership team.
- Identified inconsistent asset naming, initiating a data standardisation project that was rolled out across the department.
- Tested a Dekati particle generator for emissions equipment used across over 50 engine test cells; it was rejected after testing showed it was too unreliable and expensive.
- Used flow simulation to validate measurement probe positions, with an estimated 50% cut in measurement uncertainty.

Aim Higher Tuition | Tutor Sep 2022 – Jun 2025

- Tutored 40+ students in weekly groups of 4–5 in GCSE and A-Level Maths alongside university studies, giving focused individual attention to identify and resolve areas of weakness.
- Created weekly lesson plans on GCSE and A-Level Maths topics from 3 exam boards, tailored to each student's needs.

Motion Ventures | Software Automation Engineer Jul 2024 – Aug 2024

- Independently learned n8n within a week and built an automated lead-sourcing system integrating multiple APIs, replacing a manual process with a projected 50% improvement in data acquisition efficiency.
- Received a commendation from the founder for completing the project a week ahead of schedule.

Leadership Experience and Extracurriculars

Imperial College Islamic Society | Volunteers Lead Sep 2024 – Sep 2025

- Led volunteers across 4 campaigns and weekly Jummah prayer sessions, including the 3-person Freshers' Fortnight publicity team that helped over 50 freshers adjust to university life.
- Coordinated volunteers for Charity Week, which raised over £50,000 in total, and for nightly Ramadan iftars serving roughly 100 students alongside Taraweeh prayers.

Imperial Formula Racing | Head of the Formula Student Firewall Team Oct 2023 – Jul 2024

- Led a six-person team through a six-month redesign after identifying that the car's firewall breached competition regulations, delegating and coordinating design and manufacturing work.
- Produced BS 8888 engineering drawings, a bill of materials and costing, and manufactured the aluminium and PVC extension tabs in the workshop.

Believe and Do Good Campaign | Co-Lead, Refugee Drive Dec 2023 – Mar 2024

- Co-led the campaign's refugee drive in the first collaboration with Cambridge Islamic Society, working with the charity Reformation and coordinating approximately 50 volunteers.
- The campaign won Best Community Initiative at ICSMU's Volunteering and Academics Awards.

Autonomous Maze-Solving Buggy Oct 2026 – Present

- Programming a microcontroller in C, working towards an autonomous buggy that uses sensor readings to decide which way to turn and navigate a maze.

VTOL Rocket Oct 2024 – Jul 2025

- Led CAD design within a 12-member team building a vertical take-off and landing rocket targeting a 1 km apogee; delivered a prototype and technical poster on the thrust vectoring system with a four-member sub-team.

Python PDE Solver Jan 2024 – May 2024

- Built a Python/NumPy numerical model of forest-fire spread with animated heatmaps and 3D plots of temperature and fuel density; awarded a first-class grade.

Skills/Interests

Technical: Excel, PowerPoint, Power BI, Python, n8n, C, STAR-CCM+, GT-SUITE, SOLIDWORKS
Languages: English, Urdu
Other: Travelling (recent trips include Mount Toubkal in Morocco and Saudi Arabia for Umrah; planning a trip to Oman)